MOTOR VEHICLE

1. Introduction

This document presents an initial overview of a sample of motor-vehicle crash records. The supplied table contains crash date and time, location details, injury and fatality counts, contributing factors, and vehicle-type fields. The sample shown contains five records dated 10-05-2024 (as written in the source table).

Objective:

The main objective of this project is to analyze motor vehicle crash data to identify accident patterns, major contributing factors, high-risk locations, and the severity of road accidents. Using Python, Pandas, Matplotlib, and Seaborn, the project aims to transform raw crash data into meaningful insights that can help improve road safety and reduce accidents, injuries, and fatalities.

2. Dataset structure

Column group	Examples of fields	Purpose
Date and time	CRASH DATE, CRASH TIME	Identify when the crash occurred.
Geographic details	BOROUGH, ZIP CODE, LATITUDE, LONGITUDE, LOCATION	Describe where it occurred.
Road details	ON STREET NAME, CROSS STREET NAME, OFF STREET NAME	Record road or intersection information
Casualties	NUMBER OF PERSONS INJURED/KILLED; pedestrian, cyclist and motorist counts	Measure reported injuries and deaths.
Crash factors	CONTRIBUTING FACTOR VEHICLE 1 and 2	Capture reported contributing factors.
Vehicle details	VEHICLE TYPE CODE 2, 3, 4 and related vehicle fields	Describe involved vehicle types.

3.Data types to check

Field type	Suggested type / treatment
CRASH DATE	Convert to datetime after confirming the date format.
CRASH TIME	Parse as time or retain as text initially; validate values such as HH: MM.

ZIP CODE	Keep as string to preserve leading zeros and avoid treating it as a measurement.
LATITUDE / LONGITUDE	Numeric (float); check valid geographic ranges and missing values.
Injury / fatality counts	Numeric integers; inspect blanks, negative values, and implausible counts.
Borough, street, contributing factor, vehicle type	Categorical/text; standardize whitespace and labels such as “Unspecified”.

4.Conclusion

The sample provides useful fields for examining crash timing, location, casualties, and possible contributing factors. Before drawing conclusions, load the original CSV and verify its actual shape, column alignment, data types, missing values, duplicates, and date format. Do not treat the five sample rows as representative of the entire dataset. Important: This report is based only on the five sample rows and column names supplied in the prompt. No full-dataset totals or statistical conclusions have been inferred.